Assignment 2 — Solutions

FinTech 545 — Quantitative Risk Management

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1 Correlations from Mismatched Histories

1.1 Parts (a) through (c)

Table 1: Days each pair is jointly observed

	A	B	C	D	IDX
A	242	242	221	93	242
B	242	242	221	93	242
C	221	221	228	83	228
D	93	93	83	96	96
IDX	242	242	228	96	250

All five series trade on 81 of the 250 days, or 32%.

Part (a). The complete case matrix is guaranteed PSD. It is an ordinary sample correlation matrix computed on one set of rows, so for any weight vector w the quantity $w'Cw$ is the sample variance of a real portfolio on those rows and cannot be negative. The pairwise matrix carries no such guarantee. Each entry is estimated on a different set of days, per Table 1, so the entries are not required to be mutually consistent. A set of correlations that no single data set could have produced is exactly what a negative eigenvalue is.

Part (b). If `IDX` were an exact linear combination of the four stocks, the true matrix would be singular, with a zero eigenvalue in the direction of the portfolio long `IDX` and short the basket. The tracking error is small, so the smallest eigenvalue is close to zero rather than equal to it. The share of `IDX` variance left unexplained by the basket is roughly the size of that eigenvalue, and a tracking error that is a few percent of the index volatility puts it well under 0.01.

Sampling noise in a correlation estimated on 90 days is on the order of $1/\sqrt{90} \approx 0.1$. Inconsistencies of that size across entries are more than enough to push an eigenvalue of 0.001 negative. This data set is badly exposed. A well conditioned matrix, with a smallest eigenvalue of 0.3, would likely survive the same pairwise estimation.

Part (c). The four entries involving `D` are the least reliable. They rest on 83 to 96 days against 221 to 242 for the others. They are also estimated on a different window, the last 100 days only, so they are inconsistent with the rest of the matrix by construction, not just noisy. The complete case matrix as a whole is less reliable than the pairwise entries built on 220 or more days, because every one of its entries rests on 81.

1.2 Part (d)

Table 2: Complete case correlation, 81 days

	A	B	C	D	IDX
A	1.0000	0.3967	0.2466	0.4136	0.8714
B	0.3967	1.0000	0.3169	0.2361	0.6709
C	0.2466	0.3169	1.0000	0.1852	0.5648
D	0.4136	0.2361	0.1852	1.0000	0.5719
IDX	0.8714	0.6709	0.5648	0.5719	1.0000

Table 3: Pairwise correlation

	A	B	C	D	IDX
A	1.0000	0.4686	0.4309	0.4384	0.8696
B	0.4686	1.0000	0.4630	0.2767	0.7329
C	0.4309	0.4630	1.0000	0.1846	0.7136
D	0.4384	0.2767	0.1846	1.0000	0.6057
IDX	0.8696	0.7329	0.7136	0.6057	1.0000

Table 4: Eigenvalues, smallest first

Matrix	λ_1	λ_2	λ_3	λ_4	λ_5
Complete case	0.00151	0.54835	0.69332	0.87044	2.88639
Pairwise	−0.01107	0.47142	0.52837	0.85738	3.15390

The complete case matrix is PSD, barely, with a smallest eigenvalue of 0.0015. The pairwise matrix has an eigenvalue of −0.0111. That is −0.35% of the largest eigenvalue, about thirteen orders of magnitude past anything floating point error could produce.

The Cholesky factorization succeeds on the complete case matrix. On the pairwise matrix it fails, and so does `chol_psd`. The PSD version tolerates a slightly negative pivot and zeros it, but −0.011 is far past its tolerance, so it tries to take the square root of a negative number.

The eigenvector of the negative eigenvalue is

$$(-0.394,\ -0.234,\ -0.267,\ -0.209,\ 0.822)$$

on `A`, `B`, `C`, `D`, and `IDX`. That is close to the tracking portfolio. Long the index, short the constituents, with each weight scaled by its volatility relative to the index.

1.3 Part (e)

The full history standard deviations are 0.01650, 0.01192, 0.01832, 0.02283, and 0.01228. With D the diagonal matrix of those, $\Sigma = DCD$. The tracking portfolio is $w = (-0.40, -0.30, -0.20, -0.10, 1)$, and

$$w'\Sigma_{\text{pairwise}}\, w = -1.80 \times 10^{-6}$$

The pairwise matrix says the portfolio long the index and short its basket has a negative variance. It has no standard deviation at all.

1.4 Part (f)

Table 5: The two repairs

Repair	Min eigenvalue	$\|C - C_{\text{pw}}\|_F$	Largest change	$w'\Sigma w$
Rebonato and Jackel	2.4×10^{-16}	0.01618	0.00755	6.81×10^{-7}
Higham	-9.9×10^{-10}	0.01552	0.00704	6.79×10^{-7}

Higham lands closer to the pairwise matrix in Frobenius norm, which is what it is built to minimize. Its smallest eigenvalue is -9.9×10^{-10}. That is floating point error, and `chol_psd` handles it. The Cholesky of the repaired covariance reproduces Σ to 2×10^{-13}. Convergence of the Higham norm does not guarantee a clean eigenvalue, so we checked it rather than assumed it.

Both repairs give the tracking portfolio a variance of about 6.8×10^{-7}, a standard deviation of 0.00082. The realized tracking error on the 81 complete days is 0.00057, so the repaired number is plausible in size.

1.5 Part (g)

Week 2 defines a PSD matrix as one with $w'\Sigma w \geq 0$ for every w. A covariance matrix is PSD because $w'\Sigma w$ is the variance of the portfolio w, and a variance cannot be negative. A negative eigenvalue means there is at least one portfolio for which the matrix reports a negative variance. Part (e) found it. It is the one portfolio whose variance the pairwise estimates had to get right to within about 10^{-6}, and they did not.

1.6 Part (h)

Table 6: Higham's changes, largest first

Entry	Days	Pairwise	Higham change
A, IDX	242	0.8696	−0.00704
C, IDX	228	0.7136	−0.00477
B, IDX	242	0.7329	−0.00418
D, IDX	96	0.6057	−0.00373
A, C	221	0.4309	+0.00228
A, B	242	0.4686	+0.00200
A, D	93	0.4384	+0.00178
B, C	221	0.4630	+0.00135
C, D	83	0.1846	+0.00120
B, D	93	0.2767	+0.00106

The prediction in part (c) was wrong about where the change would go. Higham moved the four `IDX` entries most, pulling each down by 0.004 to 0.007, and nudged the stock to stock entries up. `A, IDX` rests on 242 days and moved the most. `C, D` and `B, D`, two of the thinnest entries in the matrix, moved the least.

The repair has no information about sample size. It uses the geometry of the matrix. The negative eigenvalue lives in the direction of the tracking portfolio, and the nearest PSD matrix in Frobenius norm is found by giving up correlation along that direction. Every entry that enters the tracking portfolio's variance moves in proportion to its role in that direction. The index entries carry the most weight there, so they absorb the most change. All entries are equally trusted under an unweighted norm, whether they rest on 83 days or 242.

1.7 Part (i)

The Frobenius distance between the complete case and pairwise matrices is 0.424. The repair moved the pairwise matrix by 0.016. The choice of estimator matters roughly 25 times as much as the choice of repair. Single entries differ by up to 0.18 between the two estimators, for `A, C`, against at most 0.007 from either repair.

Rebonato and Jackel against Higham is a second order decision here. Deciding which days to estimate on is the first order one.

2 A Volatility Estimate After the Regime Changed

2.1 Parts (a) through (c)

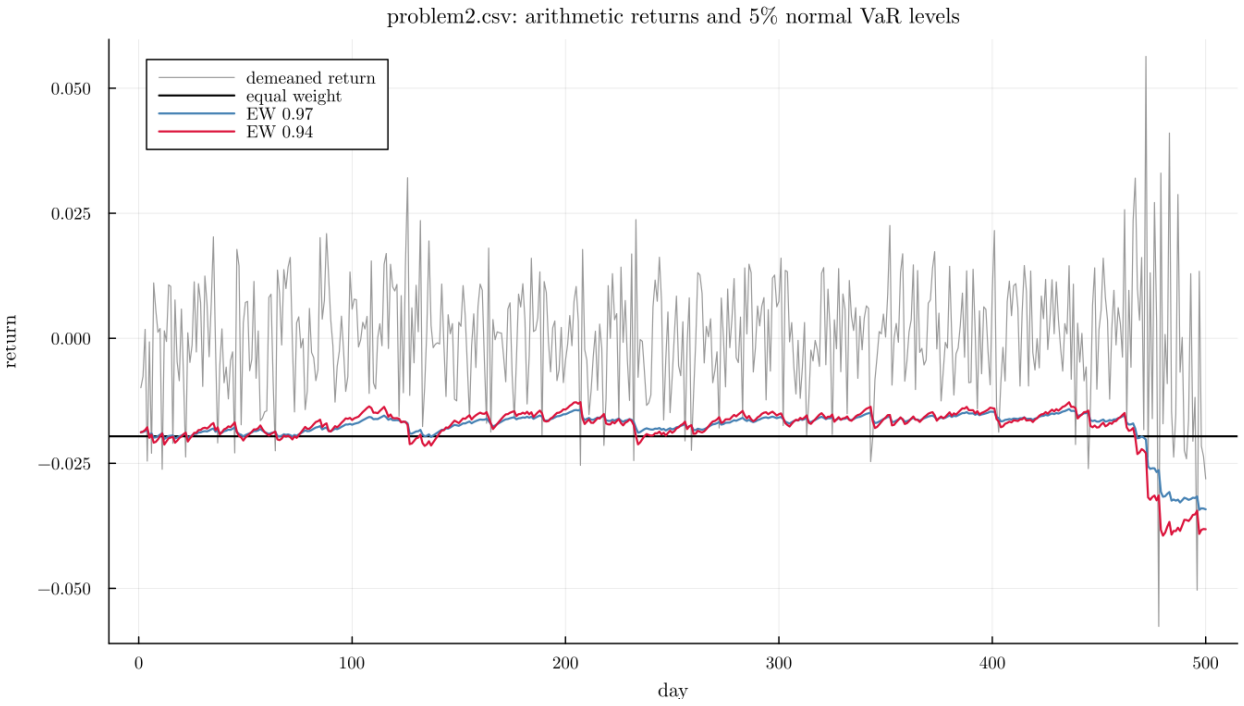


Figure 1: The demeaned returns with the 5% normal VaR level under each estimator. The exponentially weighted lines are the one day ahead estimate at each date

Figure 1 shows a long calm stretch and then a clear break in the last 40 days or so, where the moves roughly double in size.

Part (a). The ranking we expected, smallest to largest:

1. MLE t. Fit to the whole sample, it describes the calm bulk with its scale and spends ν on the stressed days. At 5% a t sits inside a normal of the same variance.
2. Normal, equal weight. Every day counts the same, so the 40 stressed days are 8% of the weight.
3. Historical. Also equal weight, but the 5% quantile of 500 days is the 25th worst day, and the stressed period supplies a disproportionate share of those. In the event, 10 of the 25 worst days come from the last 40.
4. Normal, EW $\lambda = 0.97$.
5. Normal, EW $\lambda = 0.94$. It puts the most weight on the recent period.

Part (b).

Table 7: Effective sample size, half life, and weight on the stressed period

λ	n_{eff}	Half life	Weight on last 40 days
0.94	32.3	11.2 days	91.6%
0.97	65.7	22.8 days	70.4%

The $\lambda = 0.94$ estimator is almost entirely built on the stressed period. The $\lambda = 0.97$ estimator still puts 30% of its weight on calm days, so we expect it to sit further below the stressed volatility.

Part (c). The full sample has a standard deviation of 0.01191, skewness of -0.156, and excess kurtosis of 2.13. The kurtosis rules out a single normal regime. The data looks like a mixture, and a mixture of normals with different variances has excess kurtosis even when each piece has none.

2.2 Part (d)

Table 8: One day 5% VaR on $1,000,000, sorted

Method	VaR (return)	VaR ($)
MLE t	0.0189	18,872
Normal, equal weight	0.0196	19,594
Historical	0.0206	20,574
Normal, EW 0.97	0.0341	34,064
Normal, EW 0.94	0.0363	36,308

The fitted t has location 0.000182, scale 0.01029, and $\nu = 8.27$. The exponentially weighted standard deviations are 0.02071 at $\lambda = 0.97$ and 0.02207 at $\lambda = 0.94$.

2.3 Part (e)

The order matched. The split is the part worth reading. The three full sample methods land within $1,700 of each other, between $18,900 and $20,600. The two exponentially weighted methods land at $34,000 and $36,300. The question of whether to weight recent data moves the answer by 75%. The choice among full sample methods moves it by less than 10%.

The t coming in lowest is the same crossover we saw in Assignment 1. Its implied standard deviation is $0.01029\sqrt{8.27/6.27} = 0.0118$, matching the sample. With the same variance, the t quantile is inside the normal's at 5% and outside it from about 3% down.

2.4 Part (f)

The standard deviation over the first 460 days is 0.01018, 16.2% annualized over 255 days. Over the last 40 it is 0.02429, 38.8% annualized. Inside each regime the excess kurtosis is -0.21 and -0.12. Neither piece is fat tailed.

A mixture of two normals with those variances, in a 460 to 40 ratio, has excess kurtosis

$$\frac{3\left(p_1\sigma_1^4 + p_2\sigma_2^4\right)}{\left(p_1\sigma_1^2 + p_2\sigma_2^2\right)^2} - 3 = 2.57$$

against 2.13 in the sample. The regime change alone produces the fat tails.

The fitted t is describing the mixture. Its $\nu = 8.27$ is a statement that the pooled sample has more large moves than a normal, not a statement that returns on any given day are t distributed. It answers the question "what does a random day from the last two years look like." The question for tomorrow's VaR is "what does a day in the current regime look like," and the t is not built to answer it.

2.5 Part (g)

Table 9: Estimation noise in the exponentially weighted estimates

λ	σ	Standard error	VaR $\pm$ 1 se
0.94	0.02207	0.00274 (12.4%)	\$31,793 to \$40,823
0.97	0.02071	0.00181 (8.7%)	\$31,091 to \$37,036

The two VaRs differ by \$2,244. One standard error on the $\lambda = 0.94$ VaR is about \$4,500. The gap sits well inside the noise, and we cannot say from this sample that one λ has found a different volatility than the other. What they differ on is how fast they got there and how fast they will forget it.

My opinion: $\lambda = 0.94$ for a trading limit, because a limit is supposed to bind soon after the risk changes, and 0.94 had put 92% of its weight on the new regime by the end of the sample. $\lambda = 0.97$ or slower for a capital number, where a lower standard error and less reaction to a single week are worth more than speed.

3 When Diversification Raises VaR

3.1 Parts (a) and (b)

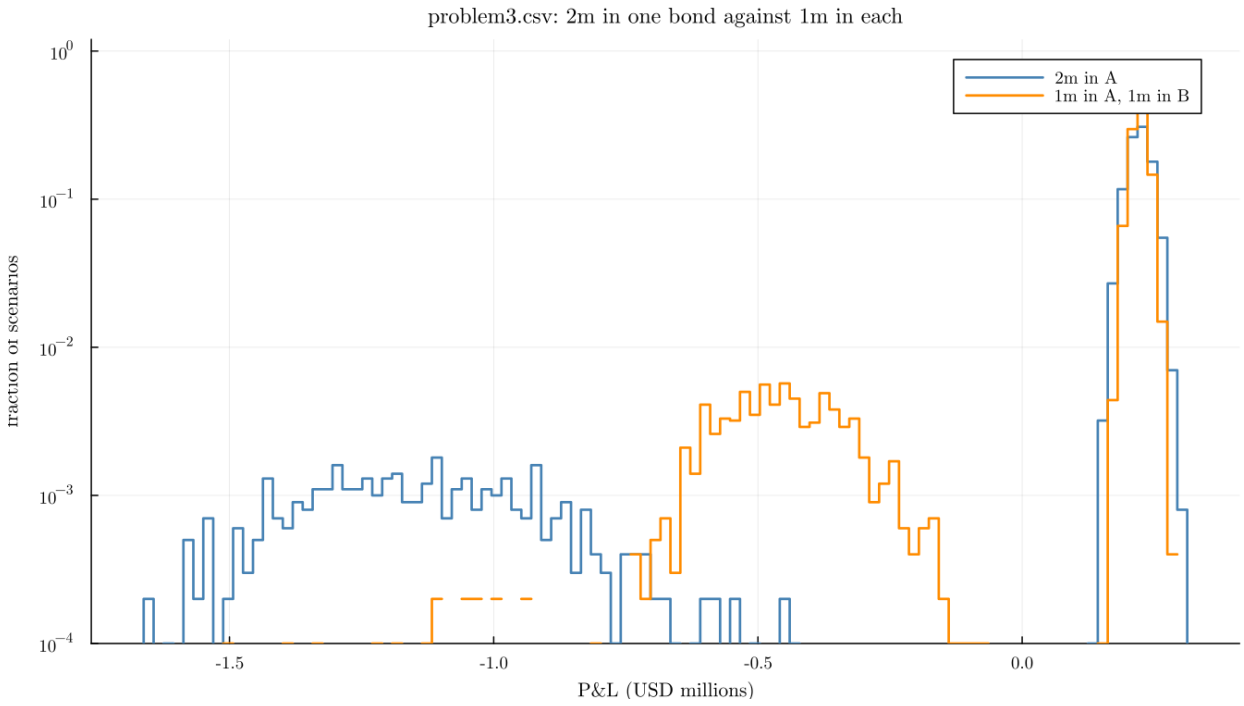


Figure 2: P&L of $2,000,000 in bond A against $1,000,000 in each bond. The vertical axis is logarithmic, so the default region is visible next to the 96% of scenarios where nothing happens

Table 10: Moments and default counts

Bond	Mean	SD	Skewness	Excess kurtosis	Losses over 20%
A	0.0837	0.1360	−4.82	22.08	406 (4.06%)
B	0.0850	0.1321	−4.92	23.07	393 (3.93%)

Part (a). A skewness near −5 and excess kurtosis above 20 say the distribution is nothing like a normal. Figure 2 shows why: 96% of the mass sits in a tight cluster around an 11% gain, and the rest sits far out at a loss of 40% to 80%. A normal fitted to these moments would put its 5% quantile somewhere in the empty space between the two clusters, where no scenario lives.

Part (b). Each bond loses more than 20% in about 4% of scenarios, below the 5% threshold. At least one of the two does in 7.82% of scenarios, and both in 0.17%.

For VaR, the concentrated position's 5% quantile falls in the no default cluster, because defaults are only 4% of scenarios. The diversified position's 5% quantile falls inside the one default region, because that region is 7.8% of scenarios. We expected diversification to raise VaR, by a lot.

For ES, the concentrated position's worst 5% contains four points of defaults, each losing roughly 55% of $2,000,000. The diversified position's worst 5% contains one bond defaulting on $1,000,000 while the other pays. We expected diversification to lower ES.

3.2 Parts (c) through (e)

Table 11: Risk of each position, in dollars

Position	VaR 5%	ES 5%	Normal VaR 5%	VaR 1%	ES 1%
$1m in A	-84,531	443,628	139,971	650,545	709,548
$1m in B	-86,011	419,587	132,277	641,248	700,805
$2m in A	-169,061	887,257	279,941	1,301,090	1,419,096
$1m in each	414,767	541,997	143,466	598,237	716,408

A negative VaR means the 5% quantile of the P&L is still a gain.

3.3 Part (f)

At 5%:

$$VaR(A) + VaR(B) = -84{,}531 - 86{,}011 = -170{,}542 < 414{,}767 = VaR(A + B)$$

$$ES(A) + ES(B) = 443{,}628 + 419{,}587 = 863{,}216 \geq 541{,}997 = ES(A + B)$$

VaR fails subadditivity and ES satisfies it. Both predictions in part (b) held.

The mechanism is the bond example from Week 4 with recovery added. Splitting the position moved probability mass across the 5% threshold. Held alone, each bond's default is a 4% event, and VaR at 5% never looks at it. Combined, "one of the two defaults" is a 7.8% event, and VaR reports it in full. ES averages over the whole 5% tail, so it sees the defaults in both cases and gives the diversified position the credit it deserves.

3.4 Part (g)

The normal VaR favours the diversified choice, $143,466 against $279,941. It gets the right answer on which choice is safer, and it gets it for the wrong reason. It is dividing the standard deviation by roughly $\sqrt{2}$, since the bonds are independent (correlation 0.002). It has no idea where the defaults sit relative to the 5% point.

Its numbers are wrong on both positions. On the concentrated position it reports a $280,000 loss where the 5% outcome is a $169,000 gain, and it reports nothing like the $887,000 average loss in the tail. The normal is averaging two clusters into one bell and reading a quantile from the gap.

3.5 Part (h)

At 1% the VaR of \$1m in A is \$650,545 and \$1m in each is \$598,237. VaR is subadditive again, $650{,}545 + 641{,}248 = 1{,}291{,}793 \geq 598{,}237$.

At 1% the threshold sits inside the default region for every position, because 4% is larger than 1%. VaR is now reading a quantile of the loss given default in each case, and diversification halves the exposure to any one default. Nothing crosses the threshold.

VaR fails only when α sits between the probability of the bad event for a single position and the probability for the combination, here between 4% and 7.8%. The failure is not a general inability to see diversification. It is blindness to whatever sits just past the chosen quantile, and whether that matters depends on where the lumps in the distribution fall relative to α.

4 Gaussian or t Copula

4.1 Part (a)

Table 12: Moments of `problem4.csv`

Series	Mean	SD	Skewness	Excess kurtosis
X1	0.00026	0.01189	−0.034	−0.123
X2	−0.00025	0.01617	−0.614	7.083
X3	0.00089	0.01002	0.074	2.854

`X1` looks normal, with skewness and excess kurtosis both near zero. `X3` is symmetric with excess kurtosis of 2.85, which points at a t.

`X2` is the one that looks out of place. A skewness of −0.61 would rule out the symmetric t by the Week 1 rule. It comes from a single day. The worst day is a return of −14.3%, 8.8 standard deviations below the mean. Without that one day the skewness is +0.056 and the excess kurtosis falls from 7.08 to 1.71. Sample skewness and kurtosis are sums of cubed and fourth powered deviations, so under fat tails one observation can dominate them. The kurtosis survives the check and the skewness does not. We fit a t to `X2`.

4.2 Part (b)

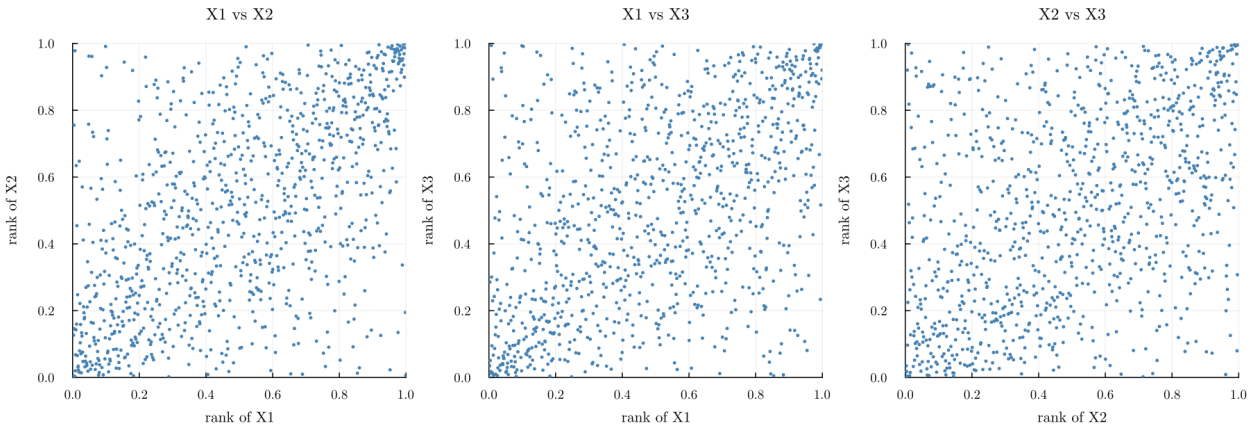


Figure 3: Every pair of series in rank space. The corners at the lower left and upper right are more crowded than the middle of the diagonal

Table 13: Days both series sit in their own tail

Pair	Worst 2.5%	Best 2.5%
X1, X2	6	8
X1, X3	7	12
X2, X3	6	9
Total	19	29

Under independence each count would be $1000 \times 0.025^2 = 0.63$, or 3.8 in total. The data shows 48. That much is just positive correlation. The question is whether it is more than the Gaussian copula implies at the same correlation, and we cannot answer that before fitting R.

Figure 3 is the stronger evidence. In each panel the points pinch together in the lower left and upper right corners, and the cloud is wider through the middle. A Gaussian copula at a correlation near 0.5 gives an ellipse that thins out evenly toward the corners. We predicted the t copula.

4.3 Part (c)

Table 14: Margin fits

Series	AICc normal	AICc t	Fitted ν	Choice
X1	−6022.54	−6020.52	7,580	normal
X2	−5408.49	−5518.18	5.27	t
X3	−6365.69	−6428.02	6.09	t

The t fitted to `X1` runs ν off to 7,580, where it is a normal, and AICc charges it about 2 for the extra parameter. The margins agree with part (a).

4.4 Part (d)

R from Kendall's τ is

$$R = \begin{bmatrix} 1 & 0.5833 & 0.5029 \\ 0.5833 & 1 & 0.4644 \\ 0.5029 & 0.4644 & 1 \end{bmatrix}$$

with a smallest eigenvalue of 0.413, so it needs no repair.

Table 15: Copula fits, $m = 1000$

Statistic	Gaussian	t
Log likelihood ℓ	350.45	452.00
Free parameters k	0	1
Fitted $\hat{\nu}$	–	3.66
AICc	−700.91	−901.99
BIC	−700.91	−897.09

$$\Delta BIC = 2(452.00 - 350.45) - \ln 1000 = 203.09 - 6.91 = 196.18$$

The profile is sharp. The log likelihood is 451.55 at $\nu = 4$, 431.56 at $\nu = 8$, and 397.92 at $\nu = 20$.

4.5 Parts (e) and (f)

Table 16: Portfolio risk, \$1,000,000 in each asset

Model	VaR 5%	ES 5%	VaR 1%	ES 1%
Historical	47,939	65,699	70,092	102,424
Gaussian copula	49,810	66,398	76,251	93,099
t copula	48,878	68,012	78,877	100,362
Change, t against Gaussian	−1.9%	+2.4%	+3.4%	+7.8%

Table 17: Joint tail days per 1,000, summed over the three pairs

Model	Worst 2.5%	Best 2.5%	Total
Data	19	29	48
Gaussian copula	14.6	14.4	29.1
t copula	24.4	23.9	48.3

4.6 Part (g)

The t copula wins by a ΔBIC of 196, far past the level of 10 read as strong evidence. AICc agrees. The prediction held.

Table 17 agrees with the fit. The Gaussian copula implies 29 joint tail days per thousand and the data has 48. The t copula implies 48.3. The Gaussian is not wrong about the correlation, since both copulas use the same R. It is wrong about how that correlation is distributed, putting too little of it in the corners.

Note that $\hat{\nu} = 3.66$ for the copula sits well below the margins' 5.27 and 6.09. The two are separate parameters. The copula's ν describes how the assets move together in the tail, and says nothing about how fat each asset's own tail is.

4.7 Part (h)

The 1% ES moves most, up 7.8%. The 5% VaR moves 1.9%, and in the other direction.

The 5% VaR of the portfolio is a point in the body of its distribution. The body is set by the margins and by R, and both copulas share both. The copulas differ in how often all three assets are extreme on the same day, which is a deep tail event. ES at 1% averages over exactly that region, so it moves. The historical 1% ES of \$102,424 sits next to the t copula's \$100,362, and the Gaussian's \$93,099 is too low.

4.8 Part (i)

The total gain $\ell_t - \ell_{\text{Gauss}}$ is 101.5. The 207 days with at least one series in its outer 5% contribute 51.0. The other 793 days contribute 50.5.

Per day, the tail days are about four times as informative, 0.247 against 0.064. There are about four times as many middle days, so the two halves carry equal weight. The criterion is not scoring the joint tail alone. It scores the whole shape of the copula. The t copula also differs from the Gaussian through the middle. At the center point, $u = (0.5, 0.5, 0.5)$, its log density is 0.38 higher than the Gaussian's, and at an off diagonal point such as $(0.2, 0.8, 0.5)$ it is 0.21 lower. The t is more concentrated along the diagonal, and half of the evidence comes from there.

That is a caution in both directions. A win on the likelihood does not mean the tail was the reason. A near tie on the likelihood could hide a real difference in the tail, since most of the sample is not in it.

4.9 Part (j)

The most correlated pair is `X1, X2`, at $\rho = 0.583$. At $\hat{\nu} = 3.66$,

$$\lambda = 2\, t_{\nu+1}\left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}}\right) = 0.322$$

Given an extreme loss in `X1`, the t copula puts the probability of an extreme loss in `X2` at about one in three, however far into the tail we go. The Gaussian copula puts it at zero in the limit, at every $\rho < 1$. The other two pairs are 0.273 and 0.253.

5 Model Based Simulation and Residual Correlation

5.1 Part (a)

Write the portfolio return for dollar holdings h_A and h_B as

$$R = (h_A\beta_A + h_B\beta_B)\, r_{MKT} + h_A\epsilon_A + h_B\epsilon_B$$

The market is independent of the errors, so

$$\mathrm{Var}(R) = (h_A\beta_A + h_B\beta_B)^2\sigma^2_{MKT} + h^2_A\sigma^2_A + h^2_B\sigma^2_B + 2h_A h_B\, \rho_\epsilon\, \sigma_A\sigma_B$$

The shortcut sets $\rho_\epsilon = 0$ and drops the last term. Stocks in the same industry should have positively correlated errors.

- `P1` has $h_A h_B > 0$. The dropped term is positive, so the shortcut understates the variance and the VaR.
- `P2` has $h_A h_B < 0$. The dropped term is negative, so the shortcut overstates the variance and the VaR.

`P2` is also close to market neutral if the betas are similar. Its variance is then mostly the error terms, so it should be the more exposed of the two.

5.2 Part (b)

Table 18: Market regressions

Stock	α (se)	β (se)	Residual SD	R^2
A	0.00021 (0.00058)	1.0353 (0.0547)	0.01290	0.418
B	0.00031 (0.00052)	0.9354 (0.0490)	0.01155	0.423

Neither α is distinguishable from zero, which fits the zero expected return assumption. We drop α from the simulation. The market's daily standard deviation is 0.01055.

The residual correlation is **0.698**. The market explains about 42% of each stock's variance, and the two residuals still move together strongly.

5.3 Parts (c) and (d)

Table 19: 5% VaR in dollars, 100,000 simulated days

Portfolio	Net beta	Full Σ_ϵ	Diagonal Σ_ϵ	Diagonal / full	Delta normal
P1	1.971	50,376	44,352	0.88	50,415
P2	0.100	15,876	28,604	1.80	15,834

5.4 Part (e)

Both moved in the predicted direction. The shortcut understates P1 by 12% and overstates P2 by 80%.

P2 is far more exposed. Its net beta is 0.10, so the market accounts for 1.2% of its variance under the full model and the residuals for 98.8%. The covariance term the shortcut drops is a large piece of what is left. P1 carries a net beta of 1.97, and the market supplies 46% of its variance, which the shortcut gets right.

The pair trade is the portfolio that exists because the two stocks share an industry. The shortcut throws away the one number that makes it a hedge, and reports it as nearly twice as risky as it is.

5.5 Part (f)

The full model VaR agrees with the delta normal VaR to within simulation error, \$50,376 against \$50,415 and \$15,876 against \$15,834. The analytic version of the full model gives exactly \$50,415 and \$15,834.

Week 5 makes the point directly. A regression of normals on normals is already inside the joint normal of $[Y\ X]$. The fitted β is $\Sigma_{YX}\Sigma_{XX}^{-1}$, and the in sample residuals are uncorrelated with the market by construction. Putting the pieces back together, $\beta\beta'\sigma^2_{MKT} + \Sigma_\epsilon$, rebuilds the sample covariance of the two stocks exactly. Fitting the model and simulating it bought us nothing a covariance matrix did not already have.

The two stop agreeing as soon as any piece is not normal. Give the errors a fitted t, or the market a fat tailed margin, and the joint distribution is no longer a multivariate normal. The model then has to be simulated through a copula, and the covariance matrix alone no longer describes it. They also stop agreeing when we change a piece on purpose, which is exactly what the diagonal Σ_ϵ did.